

SAFETY DATA SHEET



Marshal® 2.5 ULV

Version	Revision Date:	SDS Number:	Date of last issue: -
1.0	19.12.2025	50000367	Date of first issue: 19.12.2025

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Product name Marshal® 2.5 ULV

Other means of identification

Product code 50000367

1.2 Relevant identified uses of the substance or mixture and uses advised against

Use of the Sub- : Insecticide
stance/Mixture

Recommended restrictions : Use as recommended by the label.
on use

1.3 Details of the supplier of the safety data sheet

Supplier Address

FMC Chemicals (Pty) Ltd
Company Registration No.: 1988/001451/07
West End Office Park, Building C
Cnr. West Ave & Hall Street
Centurion
0014
South Africa

E-mail address: SDS-Info@fmc.com .

1.4 Emergency telephone number

For leak, fire, spill or accident emergencies, call:
South Africa: 080-001-4676 (CHEMTREC)

Medical emergency:
For any emergency or poisoning contact: Griffon Poison Infor-
mation Centre (24 hrs) - +27-(0)-82-446-8946

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

Classification (REGULATION (EC) No 1272/2008)

Acute toxicity, Category 4 H302: Harmful if swallowed.

Acute toxicity, Category 3 H331: Toxic if inhaled.

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Skin sensitisation, Sub-category 1B	H317: May cause an allergic skin reaction.
Carcinogenicity, Category 2	H351: Suspected of causing cancer.
Specific target organ toxicity - single exposure, Category 1	H370: Causes damage to organs.
Specific target organ toxicity - repeated exposure, Category 1	H372: Causes damage to organs through prolonged or repeated exposure.
Aspiration hazard, Category 1	H304: May be fatal if swallowed and enters airways.
Short-term (acute) aquatic hazard, Category 1	H400: Very toxic to aquatic life.
Long-term (chronic) aquatic hazard, Category 1	H410: Very toxic to aquatic life with long lasting effects.

2.2 Label elements

Labelling (REGULATION (EC) No 1272/2008)

Hazard pictograms :



Signal word : Danger

Hazard statements : H302 Harmful if swallowed.
H304 May be fatal if swallowed and enters airways.
H317 May cause an allergic skin reaction.
H331 Toxic if inhaled.
H351 Suspected of causing cancer.
H370 Causes damage to organs.
H372 Causes damage to organs through prolonged or repeated exposure.
H410 Very toxic to aquatic life with long lasting effects.

Precautionary statements : **Prevention:**
P260 Do not breathe mist or vapours.
P280 Wear protective gloves/ protective clothing/ eye protection/ face protection.

Response:

P301 + P310 IF SWALLOWED: Immediately call a POISON CENTER/ doctor.

P308 + P311 IF exposed or concerned: Call a POISON CENTER/ doctor.

P331 Do NOT induce vomiting.

P391 Collect spillage.

Storage:

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P403 + P233 Store in a well-ventilated place. Keep container tightly closed.

Disposal:

P501 Dispose of contents/container in accordance with local regulation.

Hazardous components which must be listed on the label:

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified carbosulfan (ISO)

Additional Labelling

EUH066 Repeated exposure may cause skin dryness or cracking. To avoid risks to human health and the environment, comply with the instructions for use.
EUH401

2.3 Other hazards

This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

SECTION 3: Composition/information on ingredients

3.2 Mixtures

Components

Chemical name	CAS-No. EC-No. Index-No. Registration number	Classification	Concentration (% w/w)
carbosulfan (ISO)	55285-14-8 259-565-9 006-084-00-5	Acute Tox. 3; H301 Acute Tox. 2; H330 Skin Sens. 1; H317 STOT SE 1; H370 (Nervous system, Bladder, Gastro- intestinal system, Blood) STOT RE 1; H372 (Nervous system, Bladder, Gastro- intestinal system, Blood) Aquatic Acute 1; H400 Aquatic Chronic 1; H410 M-Factor (Acute	25

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		aquatic toxicity): 100 M-Factor (Chronic aquatic toxicity): 10	
Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified	64742-94-5 265-198-5 649-424-00-3	Asp. Tox. 1; H304	>= 70 - < 80
carbofuran (ISO)	1563-66-2 216-353-0 006-026-00-9	Acute Tox. 2; H300 Acute Tox. 1; H330 STOT SE 1; H370 (Nervous system) STOT RE 1; H372 (Nervous system) Aquatic Acute 1; H400 Aquatic Chronic 1; H410 M-Factor (Acute aquatic toxicity): 10 M-Factor (Chronic aquatic toxicity): 100	>= 1 - <= 5

For explanation of abbreviations see section 16.

SECTION 4: First aid measures

4.1 Description of first aid measures

- General advice : Move out of dangerous area.
Consult a physician.
Show this safety data sheet to the doctor in attendance.
Symptoms of poisoning may appear several hours later.
Do not leave the victim unattended.
- Protection of first-aiders : First Aid responders should pay attention to self-protection
and use the recommended protective clothing
Avoid inhalation, ingestion and contact with skin and eyes.
If potential for exposure exists refer to Section 8 for specific
personal protective equipment.
- If inhaled : Move to fresh air.
Call a physician or poison control centre immediately.
If unconscious, place in recovery position and seek medical
advice.
If experiencing any discomfort, immediately remove from ex-
posure. Light cases: Keep person under surveillance. Get
medical attention immediately if symptoms develop. Serious
cases: Get medical attention immediately or call for an ambu-
lance.

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- In case of skin contact : If on clothes, remove clothes.
If on skin, rinse well with water.
Wash off with soap and plenty of water.
Get medical attention if irritation develops and persists.
- In case of eye contact : Flush eyes with water as a precaution.
Remove contact lenses.
Protect unharmed eye.
Keep eye wide open while rinsing.
If eye irritation persists, consult a specialist.
- If swallowed : Keep respiratory tract clear.
Do NOT induce vomiting.
Do not give milk or alcoholic beverages.
Never give anything by mouth to an unconscious person.
If symptoms persist, call a physician.
Take victim immediately to hospital.

4.2 Most important symptoms and effects, both acute and delayed

- Symptoms : Contains a cholinesterase inhibitor. Symptoms may include nausea, diarrhea, vomiting, decreased appetite, indigestion, muscle cramps, fatigue, insomnia, dizziness, headache, and lack of energy.
- On contact, the first symptoms to appear may be irritation. Symptoms of cholinesterase inhibition: nausea, headache, vomiting, cramps, weakness, blurred vision, pin-point pupils, tightness in chest, laboured breathing, nervousness, sweating, watering of eyes, drooling or frothing of mouth and nose, muscle spasms and coma.
- Risks : Harmful if swallowed or in contact with skin.
May be fatal if swallowed and enters airways.
May cause an allergic skin reaction.
Toxic if inhaled.
Suspected of causing cancer.
Causes damage to organs.
Causes damage to organs through prolonged or repeated exposure.
Repeated exposure may cause skin dryness or cracking.

4.3 Indication of any immediate medical attention and special treatment needed

- Treatment : This product contains a reversible cholinesterase inhibitor. Atropine sulfate is antidotal. Support respiration as needed with removal of secretions, maintenance of a patent airway and, if necessary, artificial ventilation. If cyanosis is absent: Adults - start treatment by giving 2 mg atropine intravenously or intramuscularly, if necessary, and repeat with 0.4 - 2.0 mg atropine at 15 minute intervals until atropinization occurs (tachycardia, flushed skin, dry mouth, mydriasis); Children

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under 12 - initial dose = 0.05 mg/kg body weight and repeat dose = 0.02 - 0.05 mg/kg body weight. Start 2-PAM at the same time, following manufacturer's recommended dosages and administration. Morphine, reserpine, phenothiazines and theophylline are probably contraindicated.

At first sign of pulmonary edema, the patient should be given supplemental oxygen and treated symptomatically. Observe patient to ensure that these symptoms do not recur as atropinization wears off. If in eyes, instill one drop of homatropine.

If any of the signs of cholinesterase inhibition occurs, call a doctor (physician), clinic or hospital immediately. Explain that the victim has been exposed to an organophosphorus insecticide.

Describe his/her condition and the extent of exposure. Immediately remove the exposed person from the area where the product is present.

ANTIDOTE: If symptoms (see subsection 4.2.) are present, administer atropine sulphate, which often is a lifesaving antidote, in large doses, TWO to FOUR mg intravenously or intramuscularly as soon as possible. Repeat at 5 to 10 minute intervals until signs of atropinisation appear and maintain full atropinisation until all organophosphate is metabolised. Relapse can occur after initial improvement. VERY CLOSE SUPERVISION OF THE PATIENT IS INDICATED FOR AT LEAST 48 HOURS, DEPENDING ON THE SEVERITY OF POISONING.

Decontamination procedures such as whole body washing, gastric lavage and administration of activated charcoal are often required.

Obidoxime chloride (Toxogonin), alternatively pralidoxime chloride (2-PAM), may be administered as an adjunct to, but not a substitute for atropine sulphate. Treatment with oxime should be maintained as long as atropine sulphate is administered.

At first sign of pulmonary oedema the patient should be given supplementary oxygen and treated symptomatically.

Much information on (acetyl)cholinesterase inhibition and its treatment can be found on the internet.

SECTION 5: Firefighting measures

5.1 Extinguishing media

Suitable extinguishing media : Dry chemical, CO₂, water spray or regular foam.
Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable extinguishing media : High volume water jet
Do not spread spilled material with high-pressure water streams.

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5.2 Special hazards arising from the substance or mixture

- Specific hazards during fire-fighting : Do not allow run-off from fire fighting to enter drains or water courses.
- Hazardous combustion products : Carbon oxides
Sulphur oxides
Nitrogen oxides (NO_x)
Hydrogen cyanide
Thermal decomposition can lead to release of toxic and irritating vapors.

5.3 Advice for firefighters

- Special protective equipment for firefighters : Firefighters should wear protective clothing and self-contained breathing apparatus.
- Further information : Collect contaminated fire extinguishing water separately. This must not be discharged into drains.
Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.
For safety reasons in case of fire, cans should be stored separately in closed containments.
Use a water spray to cool fully closed containers.

SECTION 6: Accidental release measures**6.1 Personal precautions, protective equipment and emergency procedures**

- Personal precautions : Evacuate personnel to safe areas.
Use personal protective equipment.
If it can be safely done, stop the leak.
Do not touch or walk through the spilled material.
Never return spills in original containers for re-use.
Mark the contaminated area with signs and prevent access to unauthorized personnel.
Only qualified personnel equipped with suitable protective equipment may intervene.
For disposal considerations see section 13.

6.2 Environmental precautions

- Environmental precautions : Prevent product from entering drains.
Prevent further leakage or spillage if safe to do so.
If the product contaminates rivers and lakes or drains inform respective authorities.

6.3 Methods and material for containment and cleaning up

- Methods for cleaning up : Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13).
Keep in suitable, closed containers for disposal.

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6.4 Reference to other sections

See sections: 7, 8, 11, 12 and 13.

SECTION 7: Handling and storage**7.1 Precautions for safe handling**

- | | | |
|---|---|--|
| Advice on safe handling | : | Avoid formation of aerosol.
Do not breathe vapours/dust.
Avoid exposure - obtain special instructions before use.
Avoid contact with skin and eyes.
For personal protection see section 8.
Smoking, eating and drinking should be prohibited in the application area.
Provide sufficient air exchange and/or exhaust in work rooms.
Dispose of rinse water in accordance with local and national regulations.
Persons susceptible to skin sensitisation problems or asthma, allergies, chronic or recurrent respiratory disease should not be employed in any process in which this mixture is being used. |
| Advice on protection against fire and explosion | : | Do not spray on a naked flame or any incandescent material.
Keep away from open flames, hot surfaces and sources of ignition. |
| Hygiene measures | : | Avoid contact with skin, eyes and clothing. When using do not eat or drink. When using do not smoke. Wash hands before breaks and immediately after handling the product. |

7.2 Conditions for safe storage, including any incompatibilities

- | | | |
|---|---|---|
| Requirements for storage areas and containers | : | Prevent unauthorized access. No smoking. Keep container tightly closed in a dry and well-ventilated place. Containers which are opened must be carefully resealed and kept upright to prevent leakage. Observe label precautions. Electrical installations / working materials must comply with the technological safety standards. |
| Advice on common storage | : | Do not store near acids. |
| Further information on storage stability | : | No decomposition if stored and applied as directed. |

7.3 Specific end use(s)

- | | | |
|-----------------|---|---|
| Specific use(s) | : | Registered pesticide to be used in accordance with a label approved by country-specific regulatory authorities. |
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SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Occupational Exposure Limits

Contains no substances with occupational exposure limit values.

Derived No Effect Level (DNEL) according to Regulation (EC) No. 1907/2006

Substance name	End Use	Exposure routes	Potential health effects	Value
Soybean oil, epoxidized	Workers	Inhalation	Long-term systemic effects	11.9 mg/m ³
	Workers	Inhalation	Acute systemic effects	70 mg/m ³
	Workers	Dermal	Long-term systemic effects	1.7 mg/kg bw/day
	Workers	Dermal	Acute systemic effects	10 mg/kg bw/day
	Consumers	Inhalation	Long-term systemic effects	2.8 mg/m ³
	Consumers	Inhalation	Acute systemic effects	17.5 mg/m ³
	Consumers	Dermal	Long-term systemic effects	0.800 mg/kg bw/day
	Consumers	Dermal	Acute systemic effects	5 mg/kg bw/day
	Consumers	Oral	Long-term systemic effects	0.800 mg/kg bw/day
	Consumers	Oral	Acute systemic effects	5 mg/kg bw/day

Predicted No Effect Concentration (PNEC) according to Regulation (EC) No. 1907/2006

Substance name	Environmental Compartment	Value
Soybean oil, epoxidized	Soil	6.25 mg/kg dry weight (d.w.)

8.2 Exposure controls

Personal protective equipment

Eye/face protection : Eye wash bottle with pure water
Tightly fitting safety goggles

Hand protection
Material : Wear chemical resistant gloves, such as barrier laminate, butyl rubber or nitrile rubber.

Remarks : The suitability for a specific workplace should be discussed with the producers of the protective gloves.

Skin and body protection : Impervious clothing
Choose body protection according to the amount and concentration of the dangerous substance at the work place.

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Respiratory protection : In case of mist, spray or aerosol exposure wear suitable personal respiratory protection and protective suit.

No personal respiratory protective equipment normally required.

Protective measures : Plan first aid action before beginning work with this product. Always have on hand a first-aid kit, together with proper instructions.

Ensure that eye flushing systems and safety showers are located close to the working place.
Wear suitable protective equipment.

In the context of professional plant protection use as recommended, the end user must refer to the label and the instructions for use.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Physical state	: liquid
Form	: liquid
Colour	: amber
Odour	: hydrocarbon-like
Odour Threshold	: No data available
pH	: 4.3
Melting point/freezing point	: not determined
Boiling point/boiling range	: not determined
Flash point	: > 79 °C Method: closed cup
Evaporation rate	: not determined
Upper explosion limit / Upper flammability limit	: not determined
Lower explosion limit / Lower flammability limit	: not determined
Vapour pressure	: not determined
Relative vapour density	: not determined
Relative density	:
Density	: 8.32 lb/gal
Bulk density	: No data available
Solubility(ies)	:
Water solubility	: No data available
Solubility in other solvents	: No data available
Partition coefficient: n-octanol/water	: Not available for this mixture.
Decomposition temperature	: not determined
Viscosity	:

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Viscosity, dynamic	:	No data available
Viscosity, kinematic	:	not determined
Explosive properties	:	No data available
Oxidizing properties	:	No data available

9.2 Other information

Flammability (liquids)	:	Sustains combustion
Particle size	:	Not applicable
Particle Size Distribution	:	Not applicable
Self-ignition	:	not determined

SECTION 10: Stability and reactivity

10.1 Reactivity

No decomposition if stored and applied as directed.

10.2 Chemical stability

No decomposition if stored and applied as directed.

10.3 Possibility of hazardous reactions

Hazardous reactions	:	No decomposition if stored and applied as directed.
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Vapours may form explosive mixture with air.

10.4 Conditions to avoid

Conditions to avoid	:	Heat, flames and sparks. Avoid formation of aerosol.
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10.5 Incompatible materials

Materials to avoid	:	Avoid strong acids, bases, and oxidizers
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10.6 Hazardous decomposition products

Stable under recommended storage conditions.

SECTION 11: Toxicological information

11.1 Information on toxicological effects

Acute toxicity

Harmful if swallowed or in contact with skin.
Toxic if inhaled.

Product:

Acute oral toxicity	:	LD50 (Rat): 740 mg/kg Remarks: The value is calculated
Acute inhalation toxicity	:	Acute toxicity estimate: 0.6 mg/l Exposure time: 4 h

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Test atmosphere: dust/mist
Method: Calculation method

Acute dermal toxicity : LD50 (Rabbit): > 2,000 mg/kg

Components:

carbosulfan (ISO):

Acute oral toxicity : LD50 (Rat, female): 185 mg/kg

Acute inhalation toxicity : LC50 (Rat, female): 0.15 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist

Acute dermal toxicity : LD50 (Rat): > 2,000 mg/kg

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Acute oral toxicity : LD50 (Rat, male and female): > 5,000 mg/kg
Method: OECD Test Guideline 401
Remarks: Based on data from similar materials

Acute inhalation toxicity : LC50 (Rat): > 4.688 mg/l
Exposure time: 4 h
Test atmosphere: vapour
Assessment: The substance or mixture has no acute inhalation toxicity

Acute dermal toxicity : LD50 (Rabbit): > 2,000 mg/kg
Method: OECD Test Guideline 402
Assessment: The substance or mixture has no acute dermal toxicity

carbofuran (ISO):

Acute oral toxicity : LD50 (Rat, female): 5.3 mg/kg
GLP: yes

LD50 (Rat, male): 5.2 mg/kg
GLP: yes

Acute inhalation toxicity : LC50 (Rat, female): 0.048 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Method: OECD Test Guideline 403
Symptoms: hypoactivity, Breathing difficulties, Tremors

LC50 (Rat, male and female): ca. 0.1 - 0.11 mg/l
Exposure time: 4 h
Symptoms: ataxia, Tremors
GLP: yes

Acute dermal toxicity : LD50 (Rat): > 2,000 mg/kg
Assessment: The component/mixture is minimally toxic after

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single contact with skin.

LD50 (Rabbit, female): > 2,000 mg/kg

GLP: yes

Assessment: The component/mixture is minimally toxic after single contact with skin.

Skin corrosion/irritation

Repeated exposure may cause skin dryness or cracking.

Product:

Result : slight irritation

Remarks : May cause skin irritation and/or dermatitis.

Components:

carbosulfan (ISO):

Species : Rabbit

Result : slight irritation

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Species : Rabbit

Assessment : Repeated exposure may cause skin dryness or cracking.

Result : No skin irritation

Remarks : Minimal effects that do not meet the threshold for classification.
Based on data from similar materials

carbofuran (ISO):

Species : Rabbit

Result : No skin irritation

Serious eye damage/eye irritation

Based on available data, the classification criteria are not met.

Product:

Result : slight irritation

Remarks : Vapours may cause irritation to the eyes, respiratory system and the skin.

Components:

carbosulfan (ISO):

Species : Rabbit

Result : slight irritation

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

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Species	:	Rabbit
Assessment	:	No eye irritation
Remarks	:	Minimal effects that do not meet the threshold for classification. Based on data from similar materials

carbofuran (ISO):

Species	:	Rabbit
Result	:	No eye irritation
GLP	:	no

Respiratory or skin sensitisation

Skin sensitisation

May cause an allergic skin reaction.

Respiratory sensitisation

Based on available data, the classification criteria are not met.

Product:

Result	:	Probability or evidence of low to moderate skin sensitisation rate in humans
Remarks	:	Causes sensitisation.

Components:

carbosulfan (ISO):

Test Type	:	Buehler Test
Species	:	Guinea pig
Method	:	OECD Test Guideline 406
Result	:	Not a skin sensitizer.
Test Type	:	Patch test
Species	:	Guinea pig
Result	:	May cause sensitisation by skin contact.

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Test Type	:	Maximisation Test
Species	:	Guinea pig
Result	:	Not a skin sensitizer.
Remarks	:	Based on data from similar materials

carbofuran (ISO):

Exposure routes	:	Dermal
Species	:	Guinea pig
Result	:	Not a skin sensitizer.
Test Type	:	Buehler Test
Result	:	Not a skin sensitizer.

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Germ cell mutagenicity

Based on available data, the classification criteria are not met.

Components:**carbosulfan (ISO):**

Genotoxicity in vitro	:	Test Type: reverse mutation assay Test system: Salmonella typhimurium Result: negative Test Type: reverse mutation assay Test system: Escherichia coli Result: negative Test Type: gene mutation test Test system: Chinese hamster cells Result: negative Test Type: Chromosome aberration test in vitro Test system: Chinese hamster cells Result: negative
Genotoxicity in vivo	:	Test Type: chromosome aberration assay Species: mice Result: negative

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Genotoxicity in vitro	:	Test Type: reverse mutation assay Method: OECD Test Guideline 471 Result: negative Remarks: Based on data from similar materials
Genotoxicity in vivo	:	Test Type: Bone marrow chromosome aberration Species: Rat Application Route: inhalation (vapour) Result: negative

carbofuran (ISO):

Genotoxicity in vitro	:	Test Type: Mouse lymphoma assay Metabolic activation: with and without metabolic activation Result: positive GLP: yes Test Type: Ames test Metabolic activation: with and without metabolic activation Result: negative Test Type: Ames test Metabolic activation: <** Phrase language not available: [6N] CUST - 100000000012904 **> Result: positive
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Genotoxicity in vivo : Test Type: chromosome aberration assay
Species: Mouse
Result: negative

Test Type: Micronucleus test
Species: Mouse
Result: negative

Test Type: Sex-linked Recessive Lethal Test
Species: Drosophila melanogaster (vinegar fly)
Result: negative
GLP: yes

Test Type: Cytogenetic assay
Species: Rat
Result: negative
GLP: yes

Test Type: unscheduled DNA synthesis assay
Species: Rat
Cell type: Liver cells
Result: negative
GLP: yes

Test Type: sister chromatid exchange assay
Species: Chinese hamster
Result: positive
GLP: yes

Carcinogenicity

Suspected of causing cancer.

Product:

Carcinogenicity - Assessment : Limited evidence of carcinogenicity in animal studies

Components:

carbosulfan (ISO):

Species : Mouse
Exposure time : 2 Years
NOAEL : 2.5 mg/kg bw/day
Result : negative

Species : Rat
Exposure time : 2 Years
NOAEL : 1 mg/kg bw/day
Result : negative

Carcinogenicity - Assessment : Weight of evidence does not support classification as a carcinogen

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Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Species	: Rat, male and female
Application Route	: inhalation (vapour)
Exposure time	: 12 month(s)
NOAEC	: 1.8 mg/l
Result	: negative
Remarks	: Based on data from similar materials

Carcinogenicity - Assessment	: Not classifiable as a human carcinogen.
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carbofuran (ISO):

Carcinogenicity - Assessment	: Weight of evidence does not support classification as a carcinogen
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Reproductive toxicity

Based on available data, the classification criteria are not met.

Components:**carbosulfan (ISO):**

Effects on fertility	: Test Type: Three-generation study Species: Rat Application Route: Oral General Toxicity - Parent: NOAEL: 1.2 mg/kg bw/day Fertility: NOAEL: 1.2 mg/kg bw/day Result: negative
Effects on foetal development	: Test Type: Embryo-foetal development Species: Rat Application Route: Oral General Toxicity Maternal: NOAEL: 2 mg/kg bw/day Developmental Toxicity: NOAEL: 2 Result: negative Test Type: Embryo-foetal development Species: Rabbit Application Route: Oral General Toxicity Maternal: NOAEL: 5 mg/kg bw/day Developmental Toxicity: NOAEL: 10 Result: negative
Reproductive toxicity - Assessment	: Weight of evidence does not support classification for reproductive toxicity

carbofuran (ISO):

Effects on fertility	: Species: Rat General Toxicity - Parent: NOAEL: 1.6 mg/kg bw/day General Toxicity F1: NOAEL: 1.6 mg/kg bw/day
Effects on foetal development	: Species: Rat General Toxicity Maternal: NOAEL: 1.2 mg/kg body weight

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Developmental Toxicity: NOAEL: 1.2 mg/kg bw/day

STOT - single exposure

Causes damage to organs.

Product:

Assessment : The substance or mixture is classified as specific target organ toxicant, single exposure, category 1.

Components:

carbosulfan (ISO):

Target Organs : Nervous system, Bladder, Gastro-intestinal system, Blood
Assessment : The substance or mixture is classified as specific target organ toxicant, single exposure, category 1.

carbofuran (ISO):

Target Organs : Nervous system
Assessment : Causes damage to organs.

STOT - repeated exposure

Causes damage to organs through prolonged or repeated exposure.

Product:

Assessment : The substance or mixture is classified as specific target organ toxicant, repeated exposure, category 1.

Components:

carbosulfan (ISO):

Target Organs : Nervous system, Bladder, Gastro-intestinal system, Blood
Assessment : The substance or mixture is classified as specific target organ toxicant, repeated exposure, category 1.

carbofuran (ISO):

Target Organs : Nervous system
Assessment : Causes damage to organs through prolonged or repeated exposure.

Repeated dose toxicity

Components:

carbosulfan (ISO):

Species : Rat
NOAEL : 2 mg/kg bw/day
Application Route : Oral
Exposure time : 90 days

Species : Dog

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NOAEL : 1.6 mg/kg bw/day
Application Route : Oral
Exposure time : 6 months

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Species : Rat, male and female
NOAEC : 0.9 - 1.8 mg/l
Application Route : inhalation (vapour)
Exposure time : 12 Months

carbofuran (ISO):

Species : Rat
LOAEL : 250 mg/kg
Application Route : Dermal
Exposure time : 7 d
Dose : 0, 15, 25, 50, 250 mg/kg bw/day
Target Organs : Nervous system

Aspiration toxicity

May be fatal if swallowed and enters airways.

Product:

The substance or mixture is known to cause human aspiration toxicity hazards or has to be regarded as if it causes a human aspiration toxicity hazard.

Components:

carbosulfan (ISO):

The substance does not have properties associated with aspiration hazard potential.

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

May be fatal if swallowed and enters airways.

carbofuran (ISO):

No data available

Experience with human exposure

Components:

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Skin contact : Symptoms: Repeated exposure may cause skin dryness or cracking.

Neurological effects

Components:

carbosulfan (ISO):

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Remarks : Neurotoxicity observed in animals studies

carbofuran (ISO):

Remarks : Neurotoxicity observed in animals studies

Further information

Product:

Remarks : This substance is a reversible cholinesterase-inhibiting pesticide, which elicits symptoms in humans typical of cholinesterase inhibition including headache, light-headedness, weakness, abdominal cramps, nausea, excessive salivation, perspiration and blurred vision. More severe signs of cholinesterase inhibition include tearing, pin-point pupils, excessive respiratory secretions, cyanosis, convulsions, generalized tremor and coma. Excessive cholinesterase inhibition may result.

Remarks : Solvents may degrease the skin.

Components:

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Remarks : Vapour concentrations above recommended exposure levels are irritating to the eyes and the respiratory tract, may cause headaches and dizziness, are anaesthetic and may have other central nervous system effects. Prolonged and/or repeated skin contact with low viscosity materials may defat the skin resulting in possible irritation and dermatitis. Small amounts of liquid aspirated into the lungs during ingestion or from vomiting may cause chemical pneumonitis or pulmonary edema.

SECTION 12: Ecological information

12.1 Toxicity

Components:

carbosulfan (ISO):

Toxicity to fish : LC50 (Lepomis macrochirus (Bluegill sunfish)): 0.015 mg/l
Exposure time: 96 h

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 0.0015 mg/l
Exposure time: 48 h

Toxicity to algae/aquatic plants : EC50 (Pseudokirchneriella subcapitata (microalgae)): > 20 mg/l
Exposure time: 96 h

M-Factor (Acute aquatic toxicity) : 100

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Toxicity to fish (Chronic toxicity) : NOEC: 0.00828 mg/l
Exposure time: 21 d
Species: Pimephales promelas (fathead minnow)

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC: 0.0032 mg/l
Exposure time: 21 d
Species: Daphnia magna (Water flea)

M-Factor (Chronic aquatic toxicity) : 10

Toxicity to terrestrial organisms : 1.035 µg/bee
Species: Apis mellifera (bees)
Remarks: Oral

0.18 µg/bee
Species: Apis mellifera (bees)
Remarks: Contact

LD50: 10 mg/kg
Species: Anas platyrhynchos (Mallard duck)

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Toxicity to fish : LL50 (Oncorhynchus mykiss (rainbow trout)): 2 - 5 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203

Toxicity to daphnia and other aquatic invertebrates : EL50 (Daphnia magna (Water flea)): 1.4 mg/l
Exposure time: 48 h
Method: OECD Test Guideline 202

Toxicity to algae/aquatic plants : EL50 (Pseudokirchneriella subcapitata (green algae)): 1 - 3 mg/l
Exposure time: 24 h
Method: OECD Test Guideline 201

Toxicity to microorganisms : LL50 (Tetrahymena pyriformis): 677.9 mg/l
Exposure time: 72 h
Test Type: Growth inhibition

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : EL50: 0.89 mg/l
Exposure time: 21 d
Species: Daphnia magna (Water flea)
Method: OECD Test Guideline 211

carbofuran (ISO):

Toxicity to fish : LC50 (Lepomis macrochirus (Bluegill sunfish)): 0.12 - 0.24 mg/l
Exposure time: 96 h

LC50 (Oncorhynchus mykiss (rainbow trout)): 0.28 - 0.36 mg/l

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Exposure time: 96 h

LC50 (*Ictalurus punctatus* (channel catfish)): 0.21 mg/l
Exposure time: 96 h

LC50 (*Menidia menidia* (Atlantic silverside)): 0.035 mg/l
Exposure time: 96 h

Toxicity to daphnia and other aquatic invertebrates : EC50 (*Daphnia magna* (Water flea)): 0.038 mg/l
Exposure time: 48 h
Test Type: static test

LC50 (<** Phrase language not available: [6N] CUST - 100000000011633 **>): 0.0028 mg/l
Exposure time: 96 h
Test Type: static test

LC50 (*Hexagenia limbata*): 237 mg/l
Exposure time: 96 h
Test Type: static test

LC50 (*Penaeus duorarum* (Pink shrimp)): 0.012 mg/l
Exposure time: 96 h
Test Type: flow-through test

EC50 (*Crassostrea virginica* (atlantic oyster)): > 5 mg/l
Exposure time: 48 h
Test Type: static test

Toxicity to algae/aquatic plants : EC50 (*Pseudokirchneriella subcapitata* (green algae)): 19 mg/l
Exposure time: 72 h

M-Factor (Acute aquatic toxicity) : 10

Toxicity to fish (Chronic toxicity) : NOEC: 0.022 mg/l
Exposure time: 21 d
Species: *Oncorhynchus mykiss* (rainbow trout)

EC50: 0.15 mg/l
End point: Swimming behavior
Exposure time: 14 d
Species: *Oncorhynchus mykiss* (rainbow trout)
Test Type: flow-through test

NOEC: 0.028 mg/l
End point: Growth
Exposure time: 90 d
Species: *Oncorhynchus mykiss* (rainbow trout)

NOEC: 0.0434 mg/l
End point: Growth
Exposure time: 35 d
Species: *Cyprinodon variegatus* (sheepshead minnow)

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LOEC: 0.0097 mg/l
End point: Growth
Exposure time: 28 d
Species: Cyprinodon variegatus (sheepshead minnow)

NOEC: 0.006 mg/l
End point: Growth
Exposure time: 35 d
Species: Cyprinodon variegatus (sheepshead minnow)

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC: 0.009 mg/l
Exposure time: 21 d
Species: Daphnia magna (Water flea)
Test Type: flow-through test

NOEC: 0.0004 mg/l
Exposure time: 28 d
Species: Mysis bahia (opossum shrimp)
Test Type: flow-through test

NOEC: 0.00016 mg/l
Exposure time: 7 d
Species: Ceriodaphnia dubia (water flea)

M-Factor (Chronic aquatic toxicity) : 100

Toxicity to terrestrial organisms : LD50: 8 mg/kg
Species: Colinus virginianus (Bobwhite quail)

LD50: 4 - 6.2 mg/kg
Species: Phasianus colchicus (ring-necked pheasant)

LD50: 3.5 - 4.9 mg/kg
Species: Coturnix japonica (Japanese quail)

LD50: 0.71 - 0.86 mg/kg
Species: Anas platyrhynchos (Mallard duck)

NOEL: 10 ppm
Exposure time: 14 d
End point: Reproduction Test
Species: Anas platyrhynchos (Mallard duck)

NOEL: 180 ppm
Exposure time: 98 d
End point: Reproduction Test
Species: Colinus virginianus (Bobwhite quail)

12.2 Persistence and degradability

Product:

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Biodegradability : Remarks: No data is available on the product itself.

Components:

carbosulfan (ISO):

Biodegradability : Result: Not readily biodegradable.
Biodegradation: 28 %
Exposure time: 28 d

Stability in water : Remarks: Hydrolyses readily.

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Biodegradability : Result: Readily biodegradable.
Biodegradation: 58.6 %
Exposure time: 28 d
Method: OECD Test Guideline 301F
Remarks: Based on data from similar materials

carbofuran (ISO):

Biodegradability : Result: Not readily biodegradable.

Stability in water : Remarks: Does not readily hydrolyze

12.3 Bioaccumulative potential

Product:

Bioaccumulation : Remarks: No data is available on the product itself.

Components:

carbosulfan (ISO):

Bioaccumulation : Species: Fish
Bioconcentration factor (BCF): 990
Remarks: Can accumulate in aquatic organisms.

Partition coefficient: n-octanol/water : log Pow: 5.37
pH: 8
Method: OECD Test Guideline 107

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Bioaccumulation : Remarks: The product/substance has a potential to bioaccumulate.

Partition coefficient: n-octanol/water : log Pow: 3.72
Method: QSAR

carbofuran (ISO):

Bioaccumulation : Bioconcentration factor (BCF): 11

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Partition coefficient: n-octanol/water : Remarks: No data available

12.4 Mobility in soil**Product:**

Distribution among environmental compartments : Remarks: No data is available on the product itself.

Components:**carbosulfan (ISO):**

Distribution among environmental compartments : Remarks: Slightly mobile in soils

Stability in soil :

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Distribution among environmental compartments : Remarks: Expected to partition to sediment and wastewater solids. Moderately volatile.

carbofuran (ISO):

Distribution among environmental compartments : Koc: 17 - 28 ml/g, log Koc: < 1.44
Remarks: Highly mobile in soils

Stability in soil :

12.5 Results of PBT and vPvB assessment**Product:**

Assessment : This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

12.6 Other adverse effects**Product:**

Endocrine disrupting potential : The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.
Very toxic to aquatic life with long lasting effects.

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SECTION 13: Disposal considerations

13.1 Waste treatment methods

- | | |
|------------------------|---|
| Product | : The product should not be allowed to enter drains, water courses or the soil.
Do not contaminate ponds, waterways or ditches with chemical or used container.
Send to a licensed waste management company. |
| Contaminated packaging | : Empty remaining contents.
Do not re-use empty containers.
Packaging that is not properly emptied must be disposed of as the unused product.
Empty containers should be taken to an approved waste handling site for recycling or disposal. |

SECTION 14: Transport information

14.1 UN number

- | | |
|--------|-----------|
| UNRTDG | : UN 2992 |
| IMDG | : UN 2992 |
| IATA | : UN 2992 |

14.2 UN proper shipping name

- | | |
|--------|---|
| UNRTDG | : CARBAMATE PESTICIDE, LIQUID, TOXIC
(Carbosulfan, Carbofuran) |
| IMDG | : CARBAMATE PESTICIDE, LIQUID, TOXIC
(Carbosulfan, Carbofuran) |
| IATA | : Carbamate pesticide, liquid, toxic
(Carbosulfan, Carbofuran) |

14.3 Transport hazard class(es)

- | | Class | Subsidiary risks |
|--------|-------|------------------|
| UNRTDG | : 6.1 | |
| IMDG | : 6.1 | |
| IATA | : 6.1 | |

14.4 Packing group

- | | |
|---------------|------------|
| UNRTDG | |
| Packing group | : II |
| Labels | : 6.1 |
| IMDG | |
| Packing group | : II |
| Labels | : 6.1 |
| EmS Code | : F-A, S-A |

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IATA (Cargo)

Packing instruction (cargo aircraft)	:	662
Packing instruction (LQ)	:	Y641
Packing group	:	II
Labels	:	Toxic

IATA (Passenger)

Packing instruction (passenger aircraft)	:	654
Packing instruction (LQ)	:	Y641
Packing group	:	II
Labels	:	Toxic

14.5 Environmental hazards

UNRTDG

Environmentally hazardous	:	yes
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IMDG

Marine pollutant	:	yes
------------------	---	-----

IATA (Passenger)

Environmentally hazardous	:	yes
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IATA (Cargo)

Environmentally hazardous	:	yes
---------------------------	---	-----

14.6 Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

14.7 Transport in bulk according to Annex II of Marpol and the IBC Code

Not applicable for product as supplied.

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

The components of this product are reported in the following inventories:

TCSI	:	On the inventory, or in compliance with the inventory
TSCA	:	Product contains substance(s) not listed on TSCA inventory.
AIIC	:	Not in compliance with the inventory
DSL	:	This product contains chemical substance(s) exempt from CEPA DSL Inventory requirements. It is regulated as a pesticide subject to Pest Control Products Act (PCPA) requirements. Read the PCPA label, authorized under the Pest Con-

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trol Products Act, prior to using or handling this pest control product.

ENCS	:	Not in compliance with the inventory
ISHL	:	Not in compliance with the inventory
KECI	:	On the inventory, or in compliance with the inventory
PICCS	:	Not in compliance with the inventory
IECSC	:	Not in compliance with the inventory
NZIoC	:	Not in compliance with the inventory
TECI	:	Not in compliance with the inventory

15.2 Chemical safety assessment

No Chemical Safety Assessment has been carried out for this mixture.

SECTION 16: Other information
Full text of H-Statements

H300	:	Fatal if swallowed.
H301	:	Toxic if swallowed.
H304	:	May be fatal if swallowed and enters airways.
H317	:	May cause an allergic skin reaction.
H330	:	Fatal if inhaled.
H370	:	Causes damage to organs.
H372	:	Causes damage to organs through prolonged or repeated exposure.
H400	:	Very toxic to aquatic life.
H410	:	Very toxic to aquatic life with long lasting effects.

Full text of other abbreviations

Acute Tox.	:	Acute toxicity
Aquatic Acute	:	Short-term (acute) aquatic hazard
Aquatic Chronic	:	Long-term (chronic) aquatic hazard
Asp. Tox.	:	Aspiration hazard
Skin Sens.	:	Skin sensitisation
STOT RE	:	Specific target organ toxicity - repeated exposure
STOT SE	:	Specific target organ toxicity - single exposure

ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR - Agreement concerning the International Carriage of Dangerous Goods by Road; AIIC - Australian Inventory of Industrial Chemicals; ASTM - American Society for the Testing of Materials; bw - Body weight; CLP - Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECHA - European Chemicals Agency; EC-Number - European Community number; ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration as-

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sociated with x% growth rate response; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID - Regulations concerning the International Carriage of Dangerous Goods by Rail; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; SVHC - Substance of very high concern; TCSI - Taiwan Chemical Substance Inventory; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative

Further information

Other information :

Classification of the mixture:

Acute Tox. 4	H302
Acute Tox. 3	H331
Skin Sens. 1B	H317
Carc. 2	H351
STOT SE 1	H370
STOT RE 1	H372
Asp. Tox. 1	H304
Aquatic Acute 1	H400
Aquatic Chronic 1	H410

Classification procedure:

Based on product data or assessment
Calculation method
Based on product data or assessment
Based on product data or assessment
Based on product data or assessment
Based on product data or assessment
Based on product data or assessment
Calculation method
Calculation method

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